



JDScience BTEC Health and Social Care Unit 1 Set A mark scheme

Original JDScience mark scheme. Indicative content only. Equivalent wording that meets the command word should be credited. Not an official Pearson paper.

Use this booklet after the matching practice paper. Do not issue it with the student paper.

Time: 1 hour 30 minutes · **Total marks:** 90

Question 1 · 22 marks

Aisha is 9 months old. She lives with her mother, Farah, in a first-floor flat. Farah works evenings in a supermarket. Aisha's grandmother looks after her at night. Aisha can sit without support, reach for toys and babbles "mama". She becomes distressed when Farah leaves for work and settles when her grandmother sings to her.

When Aisha is 7 years old she starts Year 3. She can ride a scooter, write her name and has two close friends. She sometimes has tantrums if she loses a game.

(a) [2 marks]

Identify two physical developments expected in infancy.

- Award 1 mark for each valid physical development, up to 2 marks.
- Examples: sitting without support; crawling/cruising; standing with help; palmar or pincer grasp; teething; rapid weight/height gain; improved head control.

(b) [4 marks]

Describe two intellectual developments shown by Aisha at 9 months.

- Award 1 mark for identifying a development and 1 mark for a linked description, twice.
- Object permanence beginning / looking for dropped toys.
- Language: babbling, turning to name, early understanding of familiar words.
- Cause and effect: reaching for toys, repeating actions that get a response.

(c) [6 marks]

Explain how attachment may affect Aisha's emotional development in infancy.

- Award marks for linked explanation applied to Aisha, not a list of theorists only.
- Secure base: Farah/grandmother as safe people so Aisha explores then returns for comfort.
- Separation protest when Farah leaves is typical if an attachment has formed.
- Consistent comfort (grandmother singing) supports trust and emotional regulation.
- Inconsistent care because Farah works evenings could cause anxiety unless grandmother is a stable secondary attachment.
- Level 3 (5–6): applied explanation with more than one idea and a consequence for Aisha.

(d) [10 marks]

Discuss how Aisha's social development is likely to have changed from 9 months to 7 years.

- Indicative content: from dyadic attachment/stranger wariness to friendships, turn-taking, school rules, wider social circle.
- Play: solitary/parallel in infancy vs cooperative play and games with rules at 7.
- Language used to negotiate; two close friends shows preference and belonging.
- Tantrums if she loses may show still-developing emotional control in social situations.
- Level 3 (8–10): balanced discussion of change, applied to Aisha, with a conclusion.

Question 2 · 22 marks

Callum is 15. He has grown 12 cm in a year, his voice has broken and he has acne. He argues with his dad about going out, spends long periods on his phone and has joined a football team. His predicted grades have dropped. He tells the school nurse he feels “not good enough” compared with friends.

(a) [2 marks]

Identify two physical changes of puberty in adolescent males.

- Award 1 mark each, up to 2.
- Examples: growth spurt; voice breaking; facial/body hair; genital development; increased muscle mass; acne linked to hormones; sperm production.

(b) [4 marks]

Outline two intellectual changes that may occur in adolescence.

- Award 1+1 for each outlined change.
- Abstract thinking / hypothesising (formal operations).
- Better planning and considering consequences, though the prefrontal cortex is still maturing so risk-taking may continue.
- Exam-focused memory and specialist vocabulary at school.

(c) [6 marks]

Explain two ways hormonal changes may affect Callum's emotional development.

- Award up to 3 marks per way: identify hormone/change, link to emotion, apply to Callum.
- Testosterone/oestrogen fluctuations → mood swings, irritability, arguments with dad.
- Body image: acne and rapid growth → self-consciousness and “not good enough”.
- Sleep and energy changes affecting mood and motivation for school.

(d) [10 marks]

Evaluate the influence of peers on Callum's social and intellectual development.

- Positive: football team → belonging, teamwork, physical activity supporting mood; peers modelling revision.
- Negative: comparison on phone → low self-esteem; possible pressure to skip work; predicted grades dropping.
- Identity: trying roles (sport vs school) as part of forming adult identity.
- Judgement: peers can support or undermine development depending on the group and adult support.
- Level 3 (8–10): both sides, applied to Callum, with a justified conclusion.

Question 3 · 22 marks

Noah and Ellis are identical twins, aged 11. They live with their father after their parents separated last year. Noah has asthma. Ellis does not. The family has moved to a smaller house near a busy road. Their father works two jobs. Both boys still attend the same primary school and see their mother every other weekend.

(a) [2 marks]

Define the nature–nurture debate in human development.

- Award 1 mark for nature (genetic/biological inheritance) and 1 mark for nurture (environment, care, experience).
- Do not award a one-word answer of “genes versus environment” without a brief sense of development being shaped by both.

(b) [4 marks]

Describe two genetic or biological factors that may affect Noah's development.

- Identical genes shared with Ellis — any difference is more likely environmental or epigenetic, but asthma can have a genetic component.
- Asthma as a biological condition affecting physical development, energy, school sport and sleep.
- Shared inheritance from parents (height potential, predisposition).

(c) [6 marks]

Explain how two environmental or economic factors may affect the twins' development.

- Busy road: air quality worsening Noah's asthma; noise affecting sleep/concentration.

- Smaller house / two jobs: less time with father, possible reduced income, fewer clubs.
- Parental separation as a major life event affecting emotional security.
- Protective: same school and regular contact with mother can support continuity.

(d) [10 marks]

Assess the likely combined effects of these factors on the twins' emotional development.

- Same genes but different experiences (asthma, possible different copings) can lead to different emotional outcomes.
- Loss, house move and less parental time may cause anxiety, anger or clinginess.
- Routine (school, weekend contact) may buffer stress.
- Noah may feel different/limited by asthma; Ellis may feel guilty or overlooked.
- Level 3: assessment of combined nature and nurture with a reasoned conclusion about likely emotional effects.

Question 4 · 24 marks

Doreen Hale is 78. She lives alone since her husband died 18 months ago. She has osteoarthritis in both knees, wears hearing aids and has started to forget appointments. She used to run the lunch club at the community centre. Her daughter wants her to consider extra support at home.

(a) [2 marks]

Identify two physical effects of ageing.

- Award 1 mark each.
- Examples: reduced mobility/joint stiffness; sensory loss (hearing/vision); slower healing; cardiovascular change; loss of muscle mass (sarcopenia).

(b) [4 marks]

Describe two intellectual changes that may occur in later adulthood.

- Mild memory lapses vs more significant forgetfulness (appointments) which may need investigation (not automatically dementia).
- Wisdom/experience and vocabulary may remain strong.
- Slower processing speed; need for more time to learn new technology.

(c) [6 marks]

Explain how Doreen's social development may be affected by bereavement and reduced mobility.

- Loss of husband → isolation, loss of daily conversation and role as a couple.
- Osteoarthritis/hearing → harder to get to lunch club, withdraws from groups.
- Loss of the lunch-club role reduces status, purpose and social contact.
- Daughter's involvement can maintain some social support if accepted.

(d) [12 marks]

Discuss how health and social care services could support Doreen's development in later adulthood.

- GP/practice nurse: pain, hearing aid review, memory assessment if appropriate.
- Physiotherapy/OT: mobility, home adaptations, falls prevention.
- Social care: homecare, befriending, day services to replace lost social role.
- Voluntary sector: lunch clubs, hearing support groups.
- Person-centred: Doreen's choice, not only her daughter's preference.
- Level 3 (7–8): a range of services linked to PIES, with a short judgement.